Retinoblastoma: proliferation up, retinal identity down

RNA-seq differential expression and pathway analysis of 19,952 genes

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The biological question

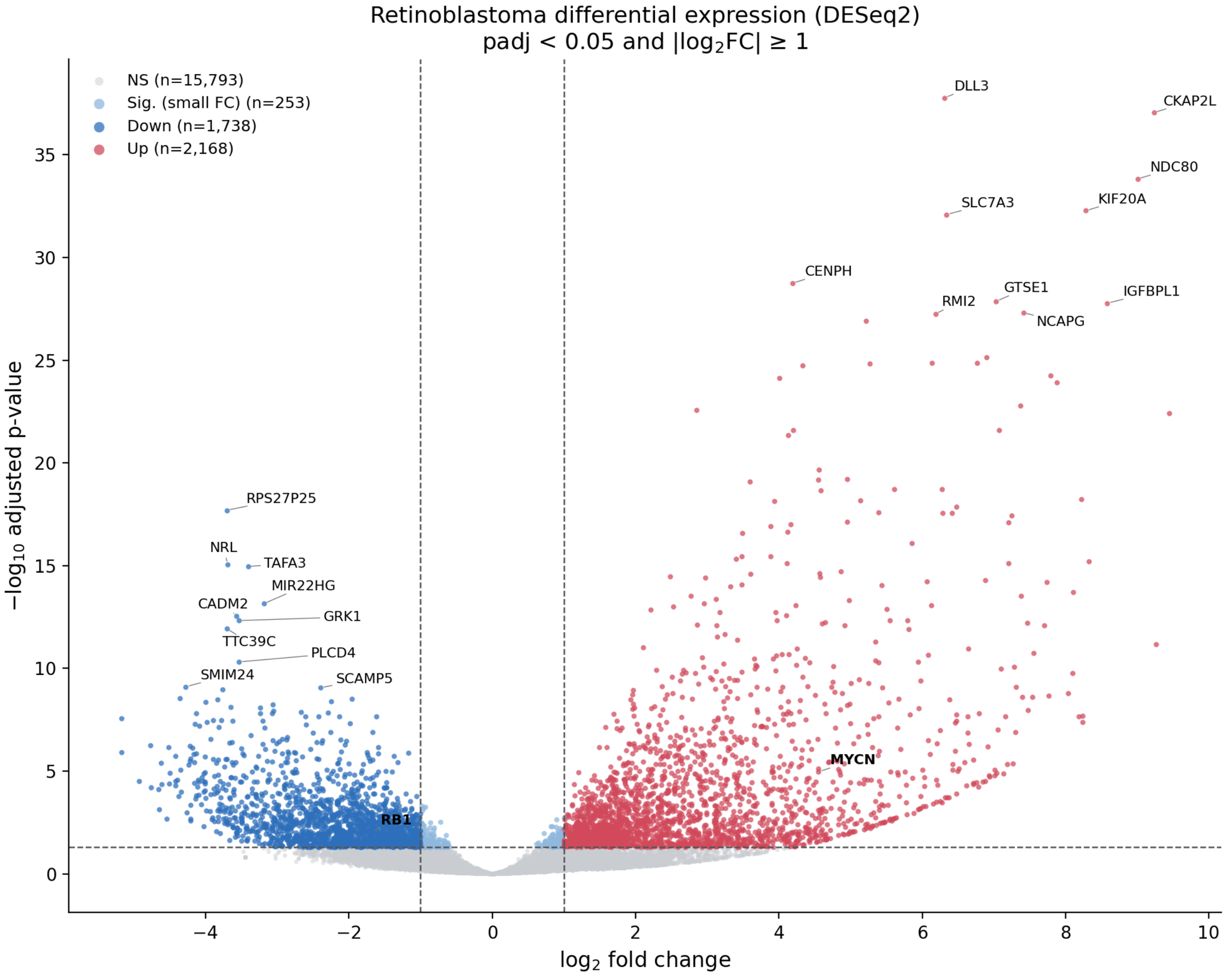
Retinoblastoma is a childhood eye cancer defined by loss of the **RB1** gene.

RB1 normally holds the **E2F** transcription factors in check. Without it, cell-cycle genes switch on.

Hypothesis

If the data are sound, cell-cycle and DNA-replication genes should be **up**, and retinal-specific genes should be **down**.

Volcano plot: 3,906 genes change strongly



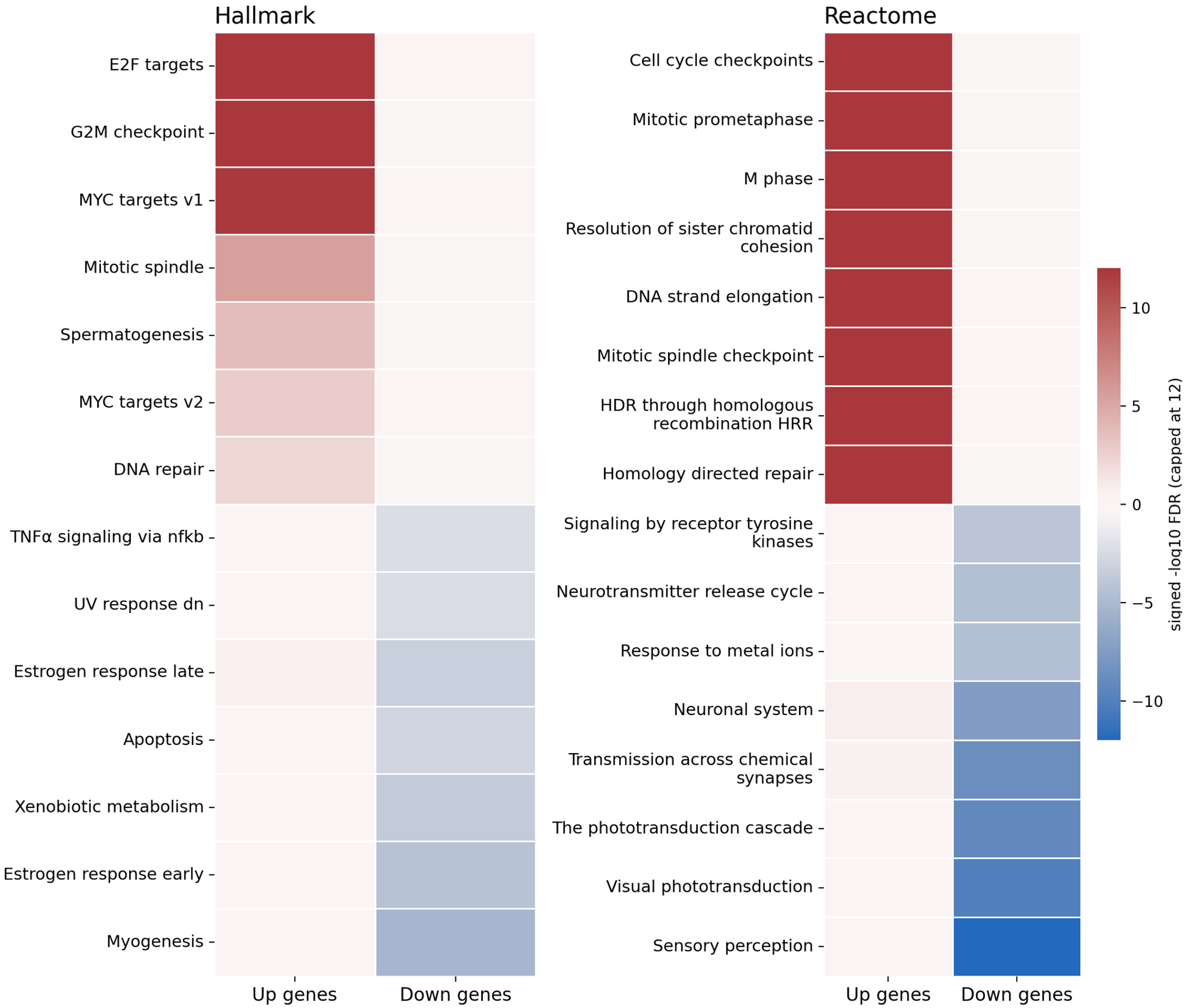
2,168 up and **1,738 down** (adjusted p < 0.05 and at least 2-fold change)

Known genes move in the expected direction

Gene	Role	log2FC	Adj. p	Direction
RB1	Tumour suppressor	-1.68	0.009	**Down**
MYCN	Oncogene	+4.55	1e-5	**Up**
MKI67	Proliferation marker	+2.77	0.031	**Up**
TOP2A	DNA replication	+3.93	0.0009	**Up**
NRL	Rod photoreceptor	-3.69	9e-16	**Down**
GRK1	Phototransduction	-3.54	5e-13	**Down**

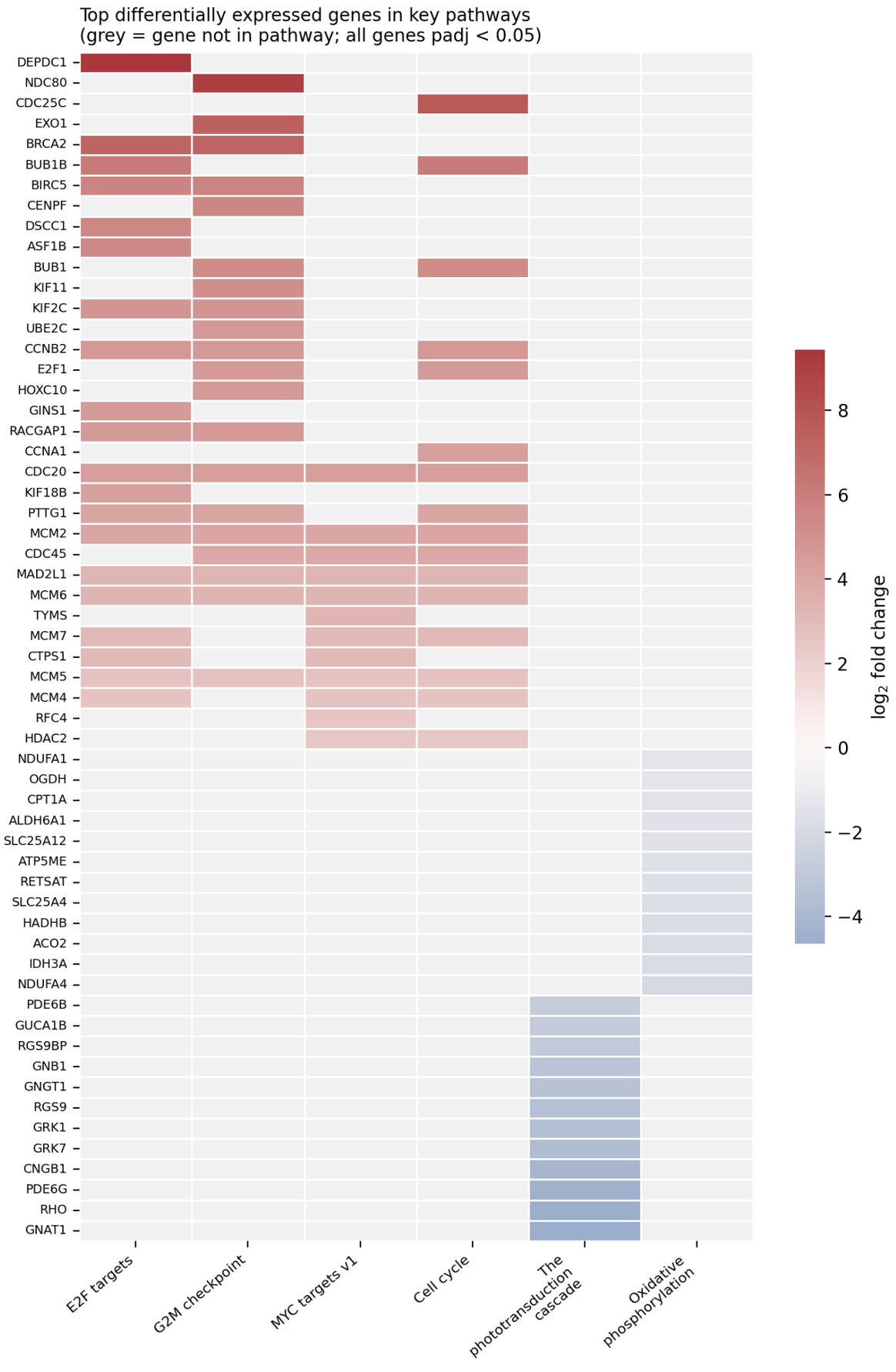
The tumour-suppressor is down, the oncogene and proliferation markers are up, and photoreceptor genes are down.

Two opposite programmes



Up: E2F targets, G2M checkpoint, MYC targets, DNA replication. **Down:** sensory perception and phototransduction.

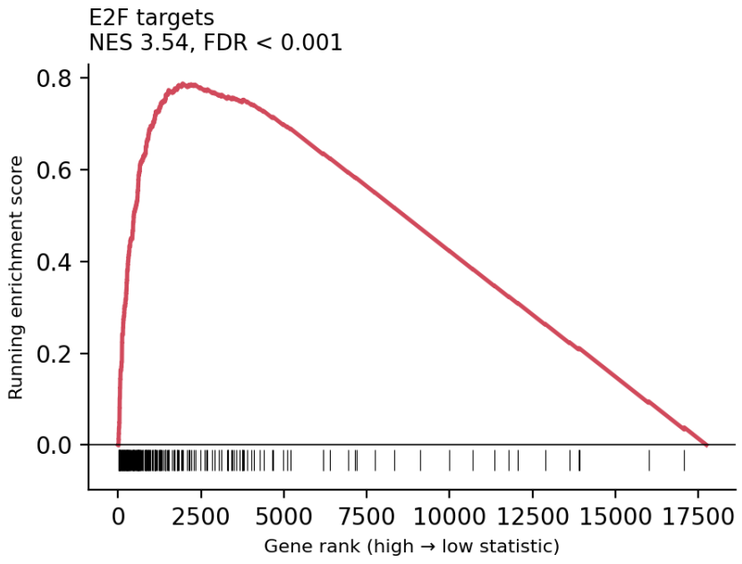
The genes behind the pathways

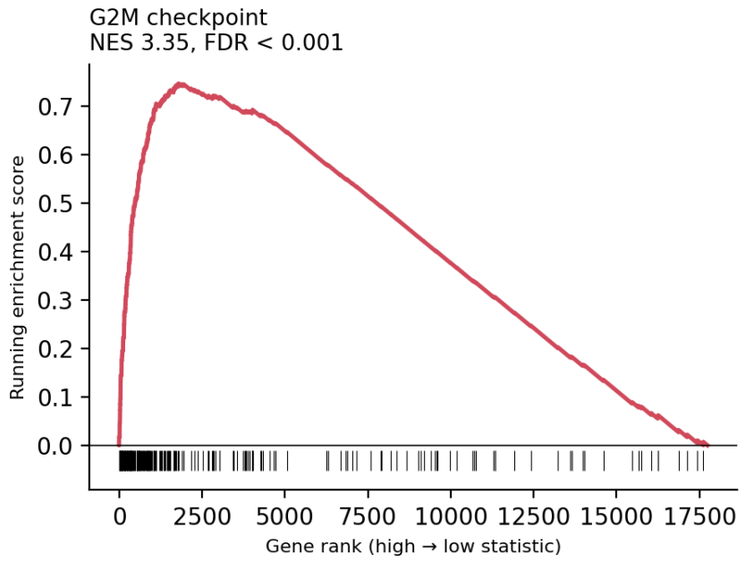


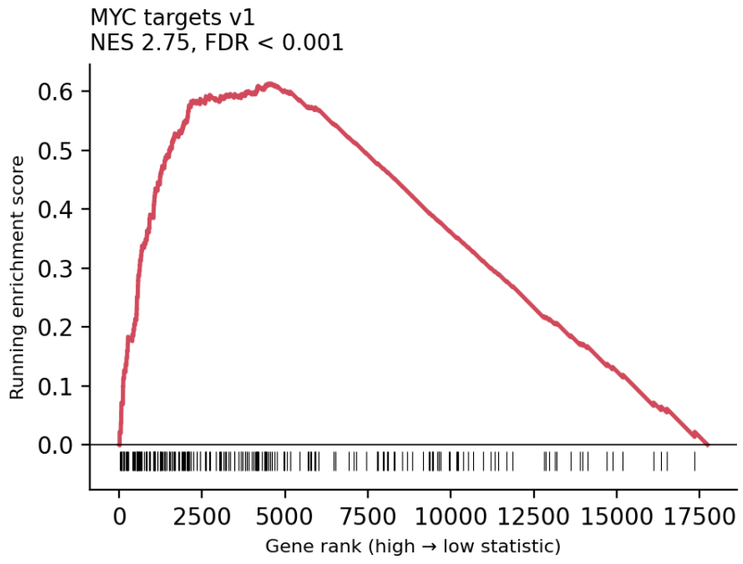
Up: NDC80, BUB1B, E2F1, MCM2 and MCM6 drive chromosome segregation and DNA replication.

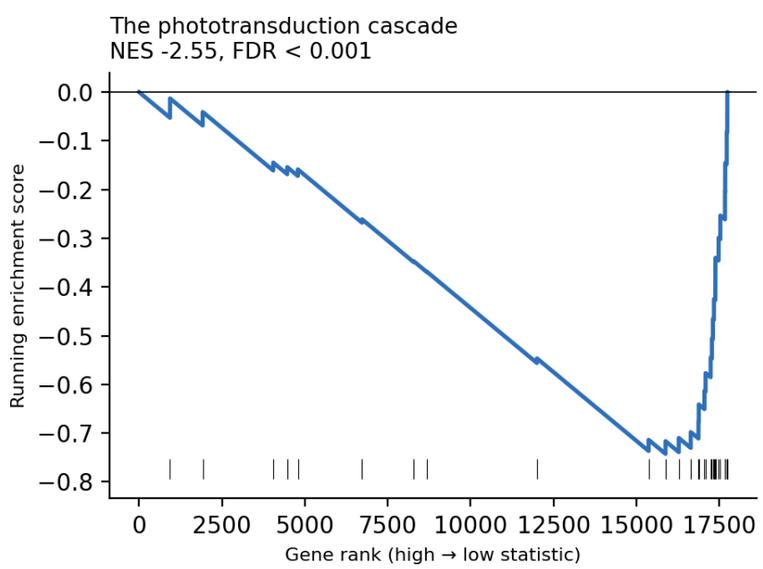
Down: RHO, GNAT1, PDE6B and GRK1 are core genes of the visual cycle in photoreceptors.

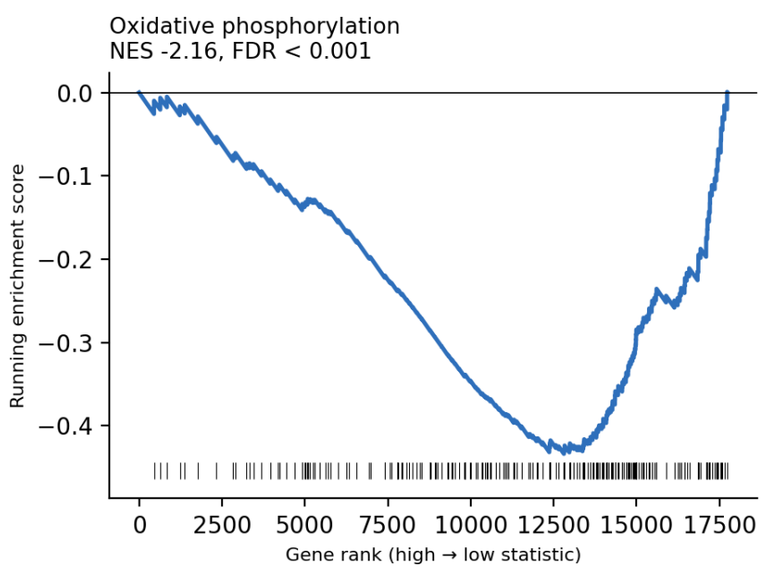
A second method agrees: GSEA

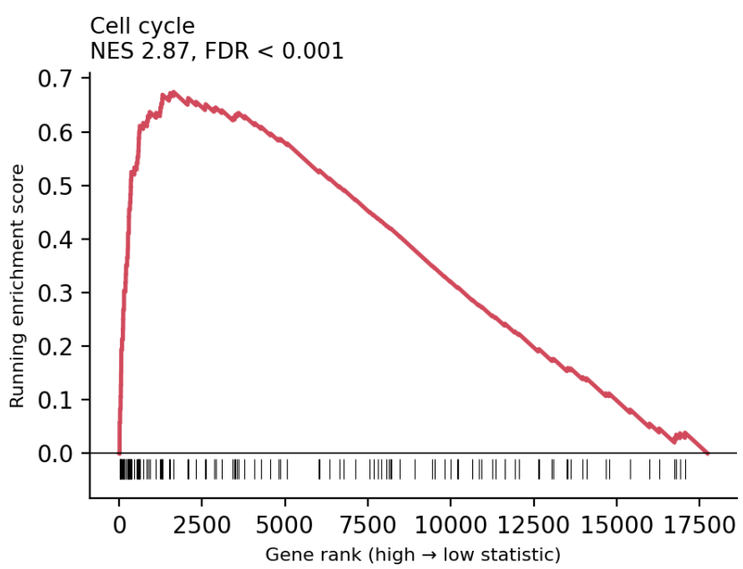












Preranked GSEA on all genes: E2F targets (NES 3.54), G2M checkpoint (3.35), phototransduction cascade (NES -2.55).

Limitations

Cell composition: the photoreceptor drop may partly reflect normal retina being rich in photoreceptors, not only tumour-intrinsic changes.

Contrast direction is assumed: positive log2FC = higher in retinoblastoma.

Annotation: 11% of tested genes had no gene symbol and were excluded from enrichment.

No sample-level plots (PCA, sample heatmap): the raw count matrix was not available.

Re-analysis: this recovers known biology and is not a new discovery.

How this was made

Data: a DESeq2 results table (57,773 genes).

Code: the Python analysis scripts and SQL queries were generated with AI assistance (Claude).

My contribution: framing the question, reviewing the outputs, checking them against known retinoblastoma biology, and interpreting the results.

Full code, SQL database and interactive volcano plot on GitHub:
[add link]